Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

SolarSense

a. Team name: **TechVerz**
b. Team leader name: **Syed Mohammed Bin Khalid**
c. Problem Statement: **Sustainability & Social Impact**

Brief about the idea

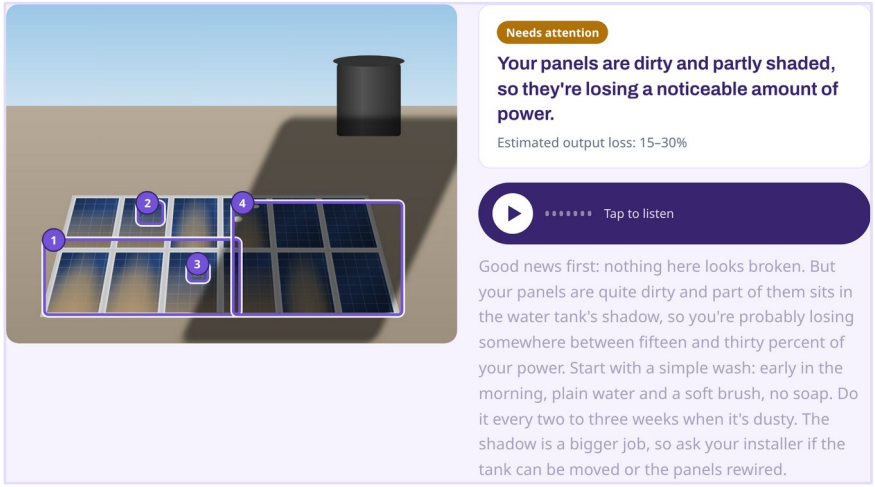
SolarSense is a talking AI solar technician.

Rooftop solar is everywhere now: 282 GW of distributed solar was added worldwide in 2025, 41% of all new solar, on homes, shops, farms and factories. But when a system underperforms or shows a fault, its owner usually can't tell what's wrong, how much it's costing them or whether it's safe.

Show it. Hear what's wrong.

The owner picks their language, then photographs the panels, inverter screen or wiring (up to 4 photos). Gemini diagnoses it like a senior technician, boxes each problem, gives the evidence and the risk, and explains it out loud and in writing in English, Urdu, Hindi, Chinese, Indonesian or Arabic. A screenshot of the inverter app shows how much money the system is losing each month, in the owner's currency, and why.

Track: Sustainability & Social Impact. **For:** home and small-business solar owners, and the installers who serve them.



Opportunities

How is it different?

Inverter apps show numbers, not answers: no verdict, no cause, no comparison with the weather, mostly English.

AI monitoring platforms exist for solar companies and need data connections, not for an owner with a phone.

SolarSense needs only a phone camera and works in the owner's own language.

How does it solve the problem?

Gemini inspects up to 4 photos together, finds dust, shade, cracks, hotspots, burnt cables and error codes, and marks them.

It speaks the diagnosis and shows it as text, with safe steps and a clear 'call a technician' when needed.

Real local sunshine turns an app screenshot into money lost per month.

USP

Show it, hear it: a voice agent you can talk back to.

6 languages incl. Urdu and Arabic, chosen on the very first screen.

Problems boxed on your own photo.

Loss shown in your own currency.

Safety first: risky jobs always go to a technician.

A PDF report the technician can act on.

List of features offered by the solution

Language first
English, Urdu, Hindi, Chinese, Indonesian or Arabic; the whole app, answers, voice and captions follow it.

Expert inspection
Up to 4 photos from the camera or gallery; each problem gets location, evidence, cause, risk and confidence.

Problems marked on the photo
Numbered boxes around each issue, with severity from minor to danger.

Voice plus text
Tap to hear the diagnosis while captions follow, with a written card for each issue.

Ask by voice
Ask a follow-up in any language; the answer looks at your photos and says what to check.

Performance check
Inverter app screenshots plus location: actual vs expected output from real local sunshine.

Money lost and why
Loss per month in your currency, with likely causes, the evidence and how to check each.

Technician report
A PDF for the installer: findings, evidence, what to do; shared on WhatsApp in one tap.

Process flow diagram or Use-case diagram

First screen: choose your language (6 options). Every text, caption and voice after this uses it.

A. Talking inspection: show the problem, hear the answer

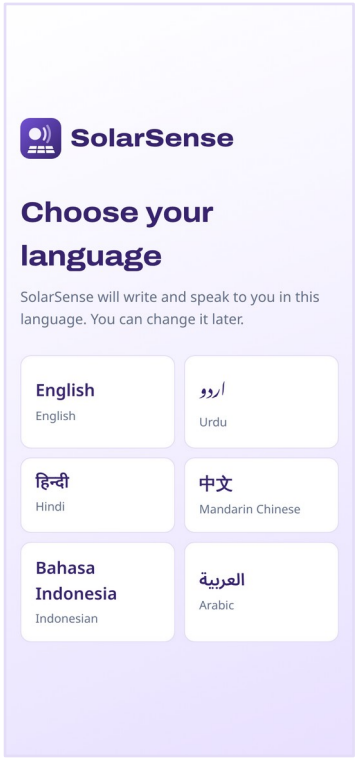
1. **Show it** — Up to 4 photos of panels, inverter screen or wiring
2. **Gemini inspects** — Finds dust, shade, cracks, hotspots, burnt cables, reads error codes
3. **Marks the photo** — Boxes around each problem, with severity and likely cause
4. **Speaks and shows** — Voice in your language, with captions and a written card for each issue
5. **Ask by voice** — Hold the mic, ask a follow-up, hear and read the answer
6. **Act** — Safe steps for you, or a PDF report for the technician on WhatsApp. Saved in Firestore

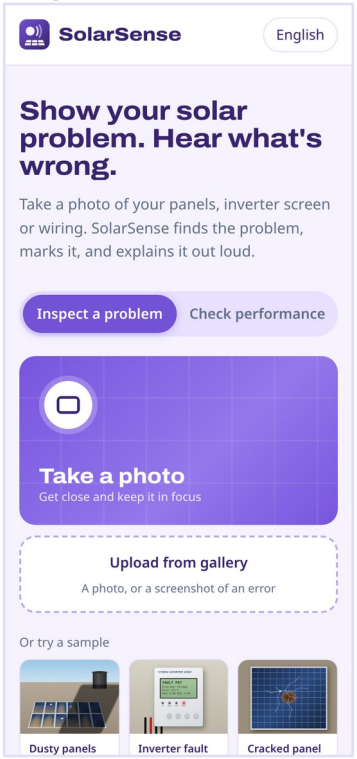
B. Performance check: is my system making what it should?

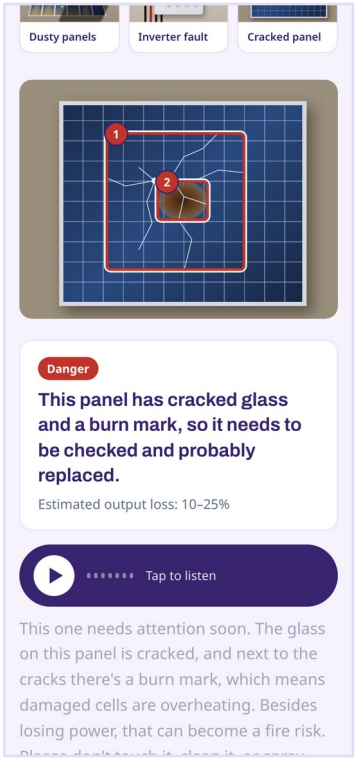
1. **Screenshot** — Inverter app screen, plus location, electricity rate and currency
2. **Gemini reads it** — Capacity, daily and monthly kWh, fault codes, any app brand
3. **Expected output** — Real sunshine at your location (Open-Meteo) × capacity × 0.78
4. **The gap in your money** — Actual vs expected, kWh and money lost per month in your currency
5. **Explains and speaks** — Likely causes with evidence and how to check, as text and voice

Gemini (AI) · App code (Firebase) · User

Wireframes/Mock diagrams of the proposed solution (optional)







Built for a phone

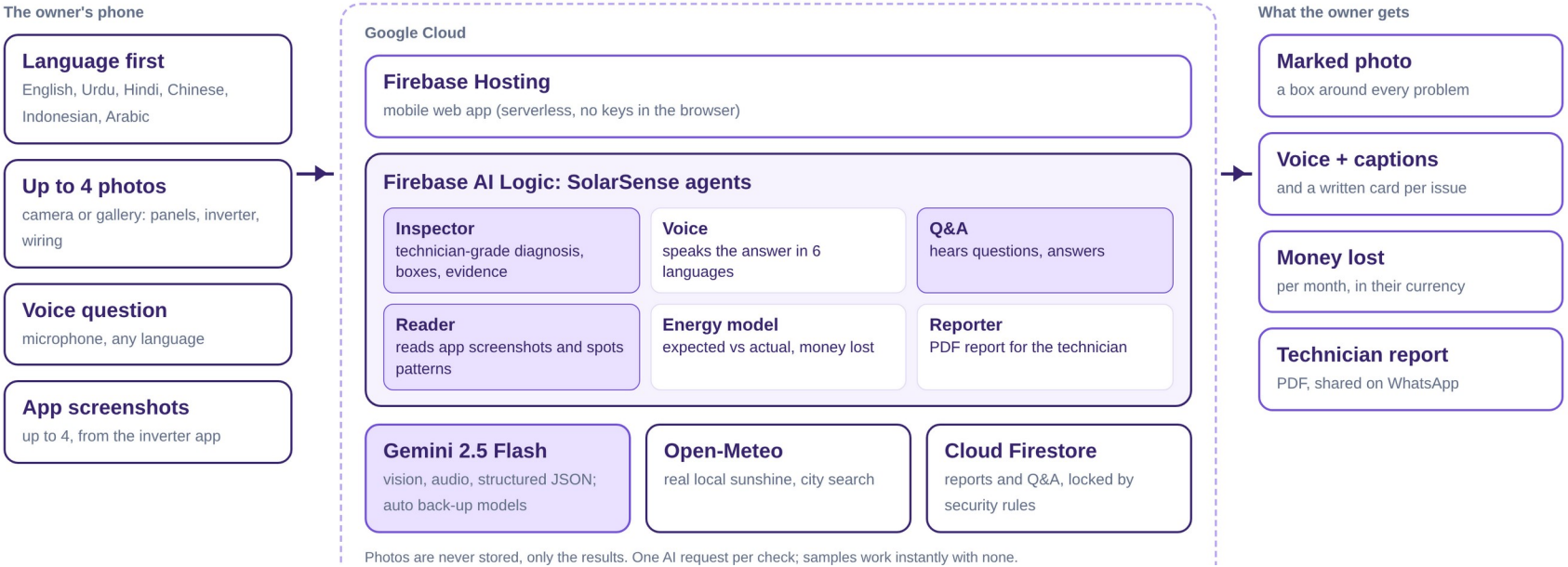
1. Pick a language. Everything after this uses it.

2. Take or upload up to 4 photos, or try a sample.

3. See the problem boxed, hear the explanation, read the captions and issue cards, then ask a follow-up by voice.

Urdu and Arabic read right to left.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Layer	Technology	How SolarSense uses it
AI: seeing	**Gemini 2.5 Flash**	Technician-grade diagnosis of up to 4 photos: boxes, severity, evidence, cause and risk as structured JSON
AI: reading	**Gemini 2.5 Flash**	Reads any inverter app screenshot (kWh, kW, fault codes) and finds the cause of low output from the pattern
AI: listening	**Gemini 2.5 Flash**	Understands typed or spoken follow-up questions in any language, and looks at the photos to answer
AI access	**Firebase AI Logic**	Calls Gemini securely from the app with no API keys in code; rotates to back-up models if one is busy
Database	**Cloud Firestore**	Reports and follow-up Q&A, protected by security rules; photos are never stored
Frontend	**Firebase Hosting**	Mobile web app with camera, microphone, voice and PDF reports; 6 languages built in
Data	**Open-Meteo, OpenStreetMap**	Real daily sunshine and city search for any location in the world

Estimated implementation cost (optional)

$0

to build and run the prototype: Firebase Spark plan, Gemini free tier, free weather and map data

< 1 cent

Gemini cost per photo check on paid usage (Gemini 2.5 Flash list prices)

1 request

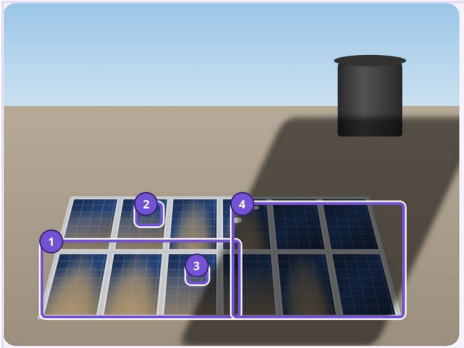
per check, however many photos; samples and maths need no AI at all

Business model

Free for owners. Installers and solar companies pay a small monthly fee to receive SolarSense reports from their customers before a visit, which saves wasted trips. Panel and inverter brands could offer it to customers as a support channel.

Estimates use Gemini 2.5 Flash list prices ($0.30 per million input tokens, $2.50 per million output tokens) and the free tiers of Firebase Hosting and Firestore. Verify against current pricing.

Snapshots of the prototype



Needs attention

Your panels are dirty and partly shaded, so they're losing a noticeable amount of power.

Estimated output loss: 15–30%

Tap to listen

Good news first: nothing here looks broken. But your panels are quite dirty and part of them sits in the water tank's shadow, so you're probably losing somewhere between fifteen and thirty percent of your power. Start with a simple wash: early in the morning, plain water and a soft brush, no soap. Do it every two to three weeks when it's dusty. The shadow is a bigger job, so ask your installer if the tank can be moved or the panels rewired.

سنگین

گرڈ کا وولٹیج زیادہ ہونے کی وجہ سے آپ کا انورٹر بند ہو گیا ہے (فالٹ F07)، اس لیے ابھی پینلز بالکل بجلی نہیں بنا رہے۔

اندازاً پیداوار میں کمی: 30–100%

سننے کے لیے دبائیں

آپ کا انورٹر فالٹ F07 کی وجہ سے خود بند ہو گیا ہے، یعنی گرڈ سے آنے والا وولٹیج بہت زیادہ ہے، آپ کی اسکرین پر دو سو اکہتر وولٹ۔ انورٹر اپنی حفاظت کے لیے بند ہوتا ہے، اس لیے ابھی آپ کے پینلز کوئی بجلی نہیں بنا رہے۔ یہ اکثر دوپہر کے وقت ہوتا ہے جہاں بہت سے گھروں پر سولر لگا ہو۔ براہ کرم انورٹر خود نہ کھولیں۔ وقت نوٹ کریں، تصویریں لیں، اور انسٹالر سے وولٹیج کی سیٹنگ اور میٹر تک کی تار چیک کروائیں۔ اگر بار بار ہو تو بجلی کمپنی کو شکایت کریں۔



You're losing about

PKR 10,114

per month. It's making 81% of what it should in this weather (202 kWh a month less).

Your system		28.4 kWh a day
Expected		35.1 kWh a day

Prototype Performance report/Benchmarking

Check	Result	How it's checked
Automated tests	4 of 4 passing	pytest: languages, money maths, sample results, API flow
End-to-end browser tests	All passing	Playwright: 6 languages, samples, camera, photo tray, PDF report, performance
Languages	6, fully translated	Interface, diagnosis, answers, captions and PDF in each language; Urdu and Arabic right to left
AI requests per check	1 (samples: 0)	Up to 4 photos or screenshots analysed in a single request
Typical answer time	about 8 to 15 s	Fast model first, automatic switch to a back-up model if one is slow
Safety rule	Always enforced	Isolators, breakers and anything electrical on a damaged system go to a technician
Free-tier resilience	6 models rotated	If one model's daily allowance is used, the next one answers

Additional Details/Future Development (if any)

Next 3 months

Accuracy test on real rooftop photos, reviewed by solar technicians

Live video mode with Gemini Live: point the camera and talk

Pilot with installers in two countries

6 to 12 months

Connect inverter cloud accounts for automatic daily checks and alerts

Installer dashboard to receive reports and book visits

Thermal photo support for hotspots

Beyond

Warranty claim packs for panel and inverter brands

Fleet view for schools, clinics and small factories

More languages and Android/iOS apps

Provide links to your:

- GitHub Public Repository: https://github.com/EmanRohail/solarsense
- Demo Video Link (3 Minutes): https://youtu.be/YOUR-VIDEO-ID
- Final Product Link: https://solarsense-6280c.web.app

The live app includes three sample photos and a sample inverter screenshot, so judges can try every feature without a solar system.

Impact

282 GW

of rooftop and other distributed solar added worldwide in 2025, 41% of all new solar

100M+

households expected to rely on rooftop solar by 2030, up from about 25 million (IEA)

6

languages, including Urdu and Arabic, so owners hear the answer in the language they think in

Who benefits

Owners: protect a big investment, cut wasted power and avoid unsafe repairs.

Installers: arrive knowing the problem, with photos and error codes in hand.

The planet: every rooftop system kept healthy is more clean energy actually produced.

Sources: IEA PVPS via TaiyangNews (2026); IEA (2022).

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Thank you

